Independent events

1 A standard six-sided die is rolled 358 times.

a If it lands on a six 12 times, find the probability that the next roll will land on a six.

b Is the outcome of the next roll independent or dependent on the outcomes of previous rolls?

2 A person runs a marathon, and then falls ill from exhaustion. Are the events independent or dependent?

3 Find the probability of drawing a green counter from a bag of 5 green counters and 6 black counters, replacing it, and drawing another green counter.

4 Two cards are to be chosen from a pack of cards numbered 1 to 6. Find the probability of drawing a 1, 2 or 3 first, and after replacing the card, drawing a 4, 5 or 6.

5 A fair die is rolled twice.

a Find the probability of rolling a 6 and a 1 in any order.

b Find the probability of rolling a 6 and then a 1.

6 A coin is tossed 600 times.

a If it lands on heads 293 times, find the probability that the next coin toss will land on heads.

b If it lands on heads once, find the probability that the next coin toss will land on heads.

c Is the outcome of the next coin toss independent or dependent on the outcomes of previous coin tosses?

7 Consider tossing a normal fair coin.

a If you have already tossed the coin 10 times, find the chance that on the next toss it will land on heads.

b If you have already tossed the coin 5 times, find the chance that the first three were a head and the last two were a tail.

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8 On a roulette table, a ball can land on one of 18 red or 18 black numbers.

a If it lands on a red number on the first go, find the probability that it will land on a red number on the second go.

b Are the successive events of twice landing on a red number dependent or independent?

c In a game of roulette, the ball has landed on a black number 6 times in a row. Is the likelihood of the ball landing on a black number in the next turn likely, unlikely or equally likely?

9 A number game uses a basket with 9 balls, all labelled with numbers from 1 to 9. Three balls are drawn at random, with replacement.

a Find the probability that the ball labelled 7 is picked exactly once.

b Find the probability that the ball labelled 7 is picked exactly twice.

c Find the probability that the ball labelled 7 is picked exactly three times.

10 Consider rolling a standard 6-sided die.

a If you roll the die six times, how many times do you expect to see a 3?

b If you roll the die once, find the probability of rolling a 3.

c You have rolled the dice three times. You did not roll a 3 in the first two rolls, but the third one was a 3. Find the exact probability of this happening.

Tree diagrams

11 A fair die is rolled and then a coin is tossed.

a Construct a tree diagram for this situation.

b Find the probability of getting an even number and a head.

c Find the probability of getting an even number, a head, or both.

12 A container holds three cards coloured red, blue and green.

a Construct a tree diagram representing all possible outcomes when two draws occur, and the card is not replaced before the next draw.

b Find the probability of drawing the blue card first.

c Find the probability of drawing a blue card in either the first or second draw.

d Find the probability of drawing at least one blue card if the cards are replaced before the next draw.

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13 A coin is tossed twice.

- **a** Construct a tree diagram to identify the sample space.
- **b** Find the probability of getting two tails.
- **c** Find the probability of getting at least one tail.

14 Christa gets to and from school by car, bus or bike.

If she goes to school by bike, she won't use the bus coming home.

Is she goes to school by bus or car, she won't cycle home.

- **a** Draw a tree diagram illustrating all possible combinations of her to and from journey to school.
- **b** State the number of possible outcomes.
- **c** If each trip is equally likely, find the probability that:
  - **i** Christa uses two different forms of transport to and from school.
  - **ii** Christa travels by car and bus.
  - **iii** Christa travels by car or bus.

15 Ivan rolled a standard die and then tossed a coin.

- **a** Construct a tree diagram to find the sample space.
- **b** List all the possible outcomes in the sample space.
- **c** Find the probability of the result including an odd number.
- **d** Find the probability that the result includes a number less than or equal to 5, and a tail.

16 Every morning Mae has toast for breakfast. Each day she either chooses honey or jam to spread on her toast, with equal chance of choosing either one.

- **a** Construct a tree diagram for three consecutive days of Mae's breakfast choices.
- **b** Find the probability that on the fourth day Mae chooses honey for her toast.
- **c** Find the probability that Mae chooses jam for her toast three days in a row.

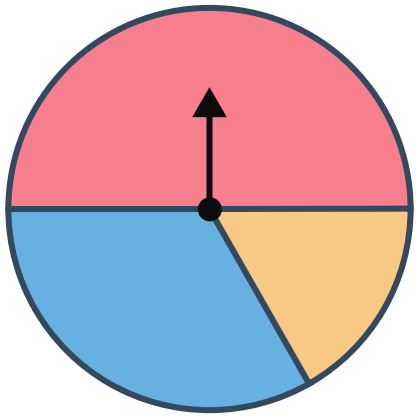
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17 A coin is tossed, then the spinner shown is spun. The blue sector is twice as big as the yellow one, and exactly half of the spinner is red.

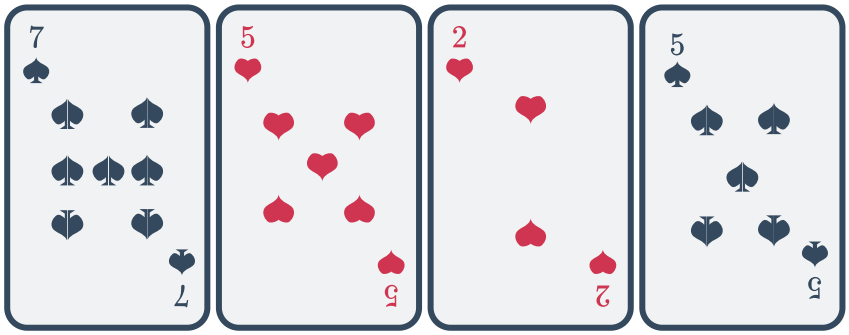
a Construct a probability tree that represents all possible outcomes.

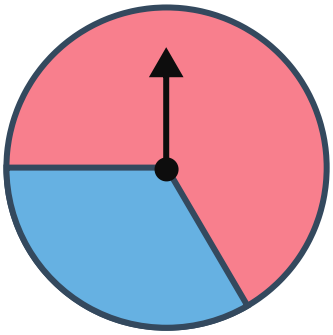
b Find the probability of throwing a heads and spinning a yellow.

c Find the probability of throwing a heads, or spinning a yellow, or both.



18 A card is selected from the four given, and the spinner is spun once:





a Construct a probability tree for this situation.

b Find the probability of choosing a 7 and blue.

c Find the probability of choosing a 7, a blue, or both.

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19 A fair coin is tossed and then the following spinner is spun:

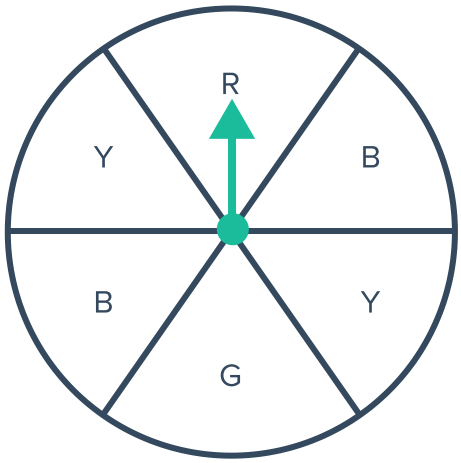
a Construct a probability tree representing the situation.

b Find the probability of getting a tail and then a yellow.

c Find the probability of getting a tail, a yellow, or both.

d Find the probability of getting a head and not getting a red.

e Find the probability of not getting a head or a red.



20 Each school day, Neil either rides his bike to school or walks. There is a 70% chance Neil will ride his bike.

a Construct a probability tree diagram showing Neil's choices for three consecutive school days.

b Find the probability that on the fourth day Neil walks to school.

c Find the probability that Neil walks to school three days in a row.

21 Robert has found that when playing chess against the computer, he wins $\frac{1}{2}$ of the time.

a Find the probability that he wins:

i Two games in a row.

ii Three games in a row.

iii At least one of two games.

iv At least one of three games.

b Which of these events does Robert have a better chance of winning?

22 Mr and Mrs Smith are starting a family. They assume that having a girl is just as likely as having a boy.

a Find the probability that the first child is a boy.

b Find the probability that the first child is a boy and the second child is a girl.

c Find the probability that the first two children are boys and the third child is a girl.

d The Smiths have three children, all boys. Find the probability that the next child will be a girl.

e What is the probability that in a four child family, the first three are boys and the fourth is a girl?

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